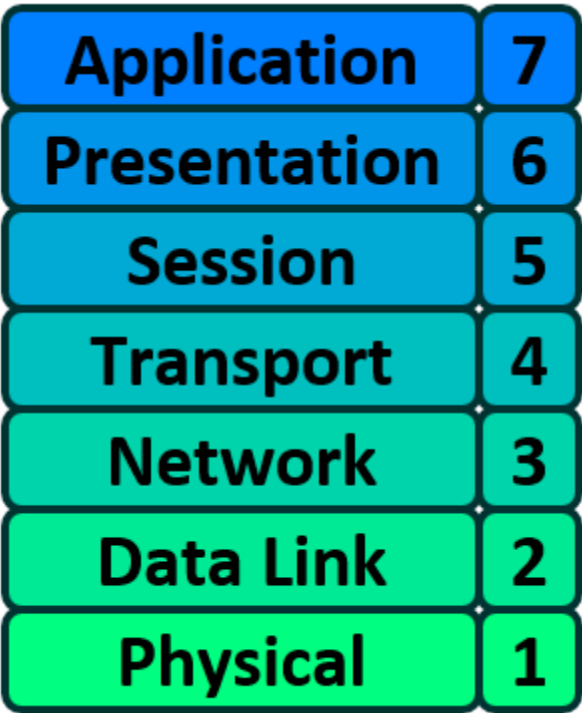


OSI Layer

- 1) Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application).



Layer No.	Layer Name	Example Device/Protocol
7	Application	HTTP, HTTPS, WhatsApp
6	Presentation	SSL/TLS (Encryption)
5	Session	NetBIOS, RPC
4	Transport	TCP, UDP
3	Network	Router, IP
2	Data Link	Switch, MAC Address
1	Physical	Ethernet Cable, Hub, Wi-Fi Signal

- 2) Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your browser, e) Detecting and correcting errors in frames.

Hint: Write your answers as 'Scenario: Layer Name (Layer Number)'.

Scenario	Layer Name (Layer Number)
a) Encrypting data before sending	Presentation Layer (Layer 6)
b) Routing packets across networks	Network Layer (Layer 3)
c) Converting data to electrical signals	Physical Layer (Layer 1)
d) Opening a web page in your browser	Application Layer (Layer 7)
e) Detecting and correcting errors in frames	Data Link Layer (Layer 2)

- 3) Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.

When I send a message on **WhatsApp**, all seven OSI layers work together:

- **Application Layer:** WhatsApp creates and sends the message.
- **Presentation Layer:** The message is encrypted using SSL/TLS.
- **Session Layer:** Maintains the connection between my phone and WhatsApp's server.
- **Transport Layer:** TCP ensures the message reaches correctly.
- **Network Layer:** IP and routers deliver the data to the correct destination.
- **Data Link Layer:** Wi-Fi or mobile network sends frames using MAC addresses.
- **Physical Layer:** The data travels as electrical, radio, or optical signals through cables or wireless communication.

- 4) Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.

Possible OSI Layers Causing the Problem

1. **Application Layer (Layer 7)** – The web server or browser may not be responding correctly.
2. **Presentation Layer (Layer 6)** – SSL/TLS certificate or encryption problem may prevent the website from loading.
3. **Session Layer (Layer 5)** – The connection between the browser and the web server may not be established properly.

Reason

Since the ping is successful, the Network Layer (Layer 3) is working correctly, meaning the IP address is reachable. The problem is most likely in the higher layers (Application, Presentation, or Session), where web pages are requested and displayed.